

Feature Engineering & Cross-Validated Model Comparison Summary

1. Engineered Features

The goal of this stage was to create additional features that capture meaningful relationships in the Adult Census Income dataset while avoiding information leakage. Seven row-level engineered features were created using only information available for the current individual. The features were integrated into the preprocessing pipeline through a FunctionTransformer, followed by numeric scaling/imputation and categorical encoding.

Engineered Feature	Type	Creation Rule	Predictive Signal
Age_Buckets	Categorical	Convert age into predefined age groups	Captures nonlinear income differences across life stages
Is_PartTime	Binary	hours-per-week < 35	Separates part-time workers from typical full-time workers
Has_CapitalGain	Binary	capital-gain > 0	Distinguishes individuals receiving any capital gain
Log_CapitalGain	Numeric	log(capital-gain + 1)	Reduces the strong skew of capital-gain values
Higher_Education	Binary	education-num >= 13	Captures the potential income advantage associated with higher education
Is_Overtime	Binary	hours-per-week > 40	Identifies individuals working beyond a standard 40-hour week
Is_Married	Binary	Based on marital-status	Captures a potentially useful socioeconomic relationship

The univariate analysis showed that the engineered features contained useful predictive information. Mutual-information scores ranked Is_Married highest among the engineered variables at approximately **0.1039**, followed by Log_CapitalGain (**0.0837**), Age_Buckets (**0.0604**), and Higher_Education (**0.0481**). Has_CapitalGain, Is_Overtime, and Is_PartTime had smaller but still measurable signals.

The feature-engineering stage was also designed to avoid target leakage. The transformations use only current-row attributes such as age, education, capital gain, marital status, and hours worked; no target encoding was introduced.

2. Cross-Validated Model Comparison

Three classification models were selected for comparison:

- **Logistic Regression**
- **Random Forest**
- **Gradient Boosting**

All models were evaluated using the same preprocessing representation and **5-fold StratifiedKfold cross-validation** with shuffling and a fixed random state. Accuracy, ROC AUC, and F1 were used as evaluation metrics. The notebook explicitly constructs the three pipelines using the same CVPreprocessor.

The primary metric for model selection was **F1**, because the project focuses on identifying the positive income class while balancing precision and recall.

The cross-validation implementation calculates mean and standard deviation for all three metrics across the five folds.

The strongest confirmed result in the notebook is:

Model	Mean F1	ROC AUC	Accuracy
Gradient Boosting	0.6811	0.9203	0.8655
Logistic Regression	0.6728	—	—
Random Forest	—	—	—

The saved notebook does contain the CV comparison and fold-score visualization, but the primary CV execution cell is saved with a null execution count. Therefore, the unavailable Random Forest/complete CV table values should **not be reconstructed or guessed** from the notebook. The subsequent statistical analysis confirms Gradient Boosting and Logistic Regression as the two highest-F1 models.

The CV analysis therefore indicates that **Gradient Boosting is the leading candidate**, while Logistic Regression provides a competitive simpler baseline.

3. Statistical Comparison

A paired t-test was performed on the five corresponding F1 fold scores of the two highest-performing models: Gradient Boosting and Logistic Regression.

Statistic	Result
Top Model	Gradient Boosting
Second Model	Logistic Regression
Gradient Boosting Mean F1	0.681052
Logistic Regression Mean F1	0.672827
Mean Difference	0.008225
Paired t-statistic	3.823011
p-value	0.018730

The p-value of **0.01873** is below the chosen significance level of 0.05, so the difference in fold-level F1 scores is statistically significant. However, the practical improvement is relatively modest: Gradient Boosting improves mean F1 by only **0.0082, or 0.82 percentage points**. The notebook therefore correctly characterizes the result as statistically significant but practically modest.

4. Engineered Feature Importance

Feature-importance analysis was performed using Gradient Boosting and Logistic Regression. The results show that Log_CapitalGain was clearly the strongest engineered feature for both models.

```
| Engineered Feature | Tree Importance | Max |Logistic Coefficient| |
|---|---|---|
| Log_CapitalGain | 0.088131 | 6.198335 |
| Higher_Education | 0.021907 | 0.090541 |
| Is_Married | 0.004252 | 0.742781 |
| Is_Overtime | 0.004216 | 0.202465 |
| Is_PartTime | 0.002263 | 0.342854 |
| Age_Buckets | 0.002093 | 0.934009 |
| Has_CapitalGain | 0.000000 | 5.811262 |
```

Log_CapitalGain was the most important engineered feature, which is sensible because capital gain is highly skewed and the logarithmic transformation provides a more useful representation of its magnitude. Higher_Education was the next strongest feature for the tree model. Is_Married and Is_Overtime had smaller contributions.

Has_CapitalGain received zero tree importance because it overlaps substantially with Log_CapitalGain. This redundancy should be considered during later tuning. The notebook also notes that the negative Logistic Regression effect of Is_PartTime is directionally sensible.

5. Feature Selection and Recommendation for Tuning

A SelectKBest experiment using mutual information was compared with the full engineered feature representation.

Feature Set	F1	ROC AUC	Accuracy	Mean Fit Time
Full features	0.6811 ± 0.0114	0.9203 ± 0.0032	0.8655 ± 0.0048	64.17 s
SelectKBest, k=50	0.6771 ± 0.0115	0.9192 ± 0.0031	0.8639 ± 0.0050	71.68 s

The full representation performed slightly better and was also faster to fit. The notebook therefore recommends **keeping the full engineered feature set** for the next stage rather than applying SelectKBest.

6. Recommended Models and Features for Tomorrow

The recommended primary model for hyperparameter tuning is **Gradient Boosting**, because it achieved the highest mean F1 of approximately **0.6811** and significantly outperformed Logistic Regression in the paired fold-level comparison. However, because the improvement over Logistic Regression is only 0.82 percentage points, Logistic Regression should remain as an important baseline during tuning.

For features, the recommendation is to **retain the complete engineered representation** for tomorrow's tuning. In particular, Log_CapitalGain and Higher_Education deserve attention because they showed the strongest engineered-feature importance. Is_Married, Is_Overtime, Is_PartTime, and Age_Buckets should also be retained because they provide additional signal at relatively low complexity. Has_CapitalGain is the least useful tree feature and is redundant with Log_CapitalGain, so it is the first candidate for removal if dimensionality or multicollinearity becomes a concern.

Final Recommendation

Tune Gradient Boosting first using the full engineered feature set, while retaining Logistic Regression as the comparison baseline. The current evidence favors Gradient Boosting on F1, but the modest 0.82-percentage-point advantage means that future improvements should be judged by both statistical reliability and practical effect size. The full feature set should be retained initially because SelectKBest reduced F1 from **0.6811 to 0.6771** while increasing average fitting time from approximately **64.2 to 71.7 seconds**.